Notes: Appel, Gormley, and Keim (JFE, 2016)

Passive Investors, Not Passive Owners

Contents

1 Big picture

- **Question.** Does greater ownership by passive institutional investors weaken governance because these investors are "lazy," or can passive investors use their voting power to improve governance and firm performance?
- **Setting.** The paper studies U.S. public firms around the annual cutoff between the Russell 1000 and Russell 2000 indexes from 1998 to 2006. The key idea is that stocks near the cutoff are similar in size, but passive funds hold much larger stakes in firms that end up near the top of the Russell 2000 than in firms near the bottom of the Russell 1000.
- **Why this matters.** Passive funds became a large and rising share of U.S. equity ownership. If they do not monitor managers, the growth of passive investing could weaken corporate governance. If they do monitor, then the rise of passive funds changes how governance works in modern capital markets.
- **Core empirical challenge.** Passive ownership is endogenous. Firms with certain growth opportunities, ownership structures, or governance arrangements may attract different types of investors, so simple correlations between passive ownership and governance are not causal.
- **Main conceptual contribution.** The paper sharply distinguishes *passive investing* from *passive ownership*. A fund may passively track an index in portfolio choice while still being an active owner through proxy voting and engagement.
- **Main empirical strategy.** Instrument ownership by passive mutual funds with an indicator for being assigned to the Russell 2000 around the Russell 1000/2000 cutoff, after flexibly controlling for end-of-May market capitalization and float-adjusted market value.
- **First-stage result.** Membership in the Russell 2000 raises passive mutual fund ownership by about 1.09 percentage points, or roughly 0.47–0.51 standard deviations, while active mutual fund ownership does not jump around the cutoff.
- **Main governance result.** Greater passive ownership is associated with more independent directors, more removal of poison pills, fewer restrictions on shareholders' ability to call special meetings, and a lower likelihood of dual-class shares.
- **Main mechanism result.** Passive ownership is associated with less support for management proposals and more support for governance-related shareholder proposals, consistent with passive investors using *voice*.
- **Activism result.** The paper does *not* find that passive ownership works by encouraging more hedge fund activism. If anything, hedge fund activism becomes less likely when passive ownership is higher.
- **Performance result.** Passive ownership is associated with improved longer-term performance, especially higher ROA once the authors control for firms that recently switched indexes. The paper also reports positive unreported evidence for Tobin's Q.
- **What passive investors do not seem to change much.** The paper finds little evidence of effects on leverage, investment, cash holdings, acquisitions, or the level and composition of executive compensation.

- **Economic takeaway.** Passive investors are not passive owners. Large, durable ownership stakes combined with proxy voting can produce meaningful governance changes even when investors do not actively trade in and out of firms.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- Passive mutual funds grew rapidly over the period that motivates the paper. The share of U.S. equity mutual fund assets in passive funds rose from roughly 9% in 1998 to 33.5% in 2014, while the share of total U.S. market capitalization held by passive mutual funds rose to above 8%.
- This rise created a first-order governance question: if a larger share of stock is held by investors who do not trade on firm-specific information, who monitors management?
- The Russell 1000/2000 design provides a useful quasi-experiment because passive ownership changes sharply at the cutoff even for firms with very similar market capitalizations.
- The setting is especially valuable because the treatment is economically meaningful. Firms at the top of the Russell 2000 receive much larger benchmark weights than firms at the bottom of the Russell 1000, so passive funds mechanically hold more of them.

Interpretation. The paper is not using a tiny or artificial source of variation. It uses a major institutional feature of U.S. equity markets that directly shapes who owns firms.

2.2 Russell 1000/2000 construction and why it creates a passive-ownership shock

- The Russell 1000 contains the largest 1,000 U.S. stocks by market capitalization; the Russell 2000 contains the next 2,000 stocks.
- Russell reconstitutes the indexes each year at the end of June using market capitalization measured on the last trading day of May.
- Within each index, stocks are value-weighted. As a result, a firm at the *top* of the Russell 2000 gets a much larger index weight than an otherwise similar firm at the *bottom* of the Russell 1000.
- Between 1998 and 2006, the average portfolio weight of the bottom 250 stocks in the Russell 1000 was 0.012%, whereas the average weight of the top 250 stocks in the Russell 2000 was 0.127%.
- Because passive funds try to minimize tracking error, they hold much more of firms that receive large benchmark weights.

Interpretation. The discontinuity is not about firm fundamentals changing abruptly at the cutoff. It is about the same dollar of passive capital being allocated very differently because of index mechanics.

2.3 Data sources and sample construction

- **Passive and active mutual fund holdings:** Thomson Reuters S12 mutual fund holdings, merged with CRSP mutual fund data.
- **Index membership and float-adjusted market value:** Russell Investments.
- **Governance and vote outcomes:** ISS / RiskMetrics (directors, governance provisions, voting results).
- **Poison pills:** SharkRepellent / FactSet.
- **Accounting outcomes:** Compustat.
- **Executive compensation:** Execucomp.
- **Hedge fund activism events:** data from Brav, Jiang, Partnoy, and Thomas, and Brav, Jiang, and Kim.

The main sample runs from 1998 to 2006.

- The sample starts in 1998 because Russell provides the relevant float-adjusted market-cap information from then onward.
- The sample ends before 2007 because Russell changed its index-construction methodology in 2007 by introducing banding rules.
- The main estimation sample uses a bandwidth of 250 firms on each side of the cutoff: the bottom 250 firms in the Russell 1000 and the top 250 firms in the Russell 2000.

Interpretation. The paper focuses tightly on firms near the cutoff so that comparison firms are similar in size and the treatment is the difference in passive ownership induced by index assignment.

2.4 How passive ownership is measured

- Funds are classified as **passive** if their names indicate index tracking or if the CRSP mutual fund database identifies them as index funds.
- Funds matched to CRSP but not identified as passive are treated as **active**.
- Unmatched funds are labeled **unclassified**.
- For each stock, the paper computes the percentage of shares outstanding held by passive, active, and unclassified mutual funds.
- Ownership is measured at the end of September following the Russell reconstitution.

The paper also cross-checks the results with broader institution-level 13F ownership data:

- "quasi-index" institutions in the Bushee classification serve as an alternative passive-ownership measure;
- the results are also robust when passive ownership is measured using the combined 13F holdings of Vanguard, State Street, and Barclays/iShares.

Interpretation. The S12 data provide a precise measure of passive mutual fund holdings, while the 13F exercises check that the findings do not depend on one ownership definition.

2.5 Governance, voting, and performance outcomes

The main outcomes are chosen to match governance issues that large passive institutions explicitly emphasized in their historical voting policies.

- **Board independence:** percentage of directors classified as independent by ISS.
- **Takeover defenses:** indicators for poison-pill removal and for fewer restrictions on shareholders' ability to call a special meeting.
- **Voting rights:** indicator for a dual-class share structure.
- **Voice mechanism:** support for management proposals and support for shareholder governance proposals.
- **Alternative channel:** indicator for hedge fund activism.
- **Performance:** ROA, and in unreported tests Tobin's Q.
- **Other corporate policies:** leverage, capital expenditures, R&D, cash holdings, acquisitions, dividends, and executive compensation.

Interpretation. The outcome set is not random. It mirrors the governance issues on which passive investors claimed they wanted to be active owners.

2.6 Descriptive facts

For the main sample around the cutoff:

- average total mutual fund ownership is 25.2% of shares outstanding;
- average passive mutual fund ownership is 3.0%;
- average active mutual fund ownership is 18.9%;
- average unclassified mutual fund ownership is 3.2%;
- independent directors make up 65.1% of boards on average;
- poison-pill removals occur in about 4% of firm-years;
- increases in shareholders' ability to call special meetings occur in about 0.6% of firm-years;
- 13% of firms have dual-class shares;
- mean support for management proposals is 84.7%, while mean support for shareholder governance proposals is 36.3%;
- hedge fund activism occurs in about 1.6% of firm-years;
- mean ROA is about 3%.

The ownership stakes of the three biggest passive institutions in the sample period—Vanguard, State Street, and Barclays/iShares—are also meaningfully higher for firms at the top of the Russell 2000.

- Their average stake is about one-third higher.
- The probability that each owns more than 5% of the firm's shares rises by roughly two-thirds.
- The probability that each is among the firm's top five shareholders rises by about 15%.

Interpretation. Even though passive mutual fund ownership averages only 3%, the governance relevance of passive investors comes from concentrated, durable voting blocs held by the largest institutions.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper's mechanism is simple.

- Passive funds generally do not influence firms through buying a large stake and then threatening to sell it if management underperforms.
- But passive funds often hold firms for long periods and have a fiduciary duty to vote their shares.
- Because they cannot easily *exit* individual firms without deviating from the benchmark, they may have stronger incentives to use *voice*.
- Their governance interventions are most likely to be effective on issues that are broad, low-cost, and scalable across many firms: board independence, takeover defenses, voting rights, and proposal voting.

This leads to three main hypotheses:

- Higher passive ownership should improve governance along dimensions emphasized in passive funds' voting policies.
- If passive investors matter through voice, proposal vote outcomes should change.
- If those governance changes are value increasing, longer-term firm performance should improve.

3.2 Baseline second-stage IV specification

The core estimating equation is:

$$Y_{it} = \alpha + \beta\, Passive\%_{it} + \sum_{n=1}^{N} \theta_n \big(\ln(Mktcap_{it})\big)^n + \gamma \ln(Float_{it}) + \delta_t + \varepsilon_{it}.$$

Where:

- Y_{it} is the governance, voting, activism, or performance outcome for firm i in reconstitution year t;
- $Passive\%_{it}$ is passive mutual fund ownership at the end of September in year t;
- $Mktcap_{it}$ is end-of-May market capitalization;
- $Float_{it}$ is Russell's float-adjusted market capitalization used when assigning portfolio weights;
- δ_t are year fixed effects.

Both Y_{it} and $Passive\%_{it}$ are scaled by their sample standard deviations in the main regressions.

Interpretation. The coefficient β tells us how many standard deviations the outcome moves when passive ownership increases by one standard deviation.

3.3 First stage

The instrument is Russell 2000 membership:

$$Passive\%_{it} = \eta + \lambda R2000_{it} + \sum_{n=1}^{N} \chi_n \big(\ln(Mktcap_{it})\big)^n + \sigma \ln(Float_{it}) + \delta_t + u_{it}.$$

Where $R2000_{it}$ equals one if firm i belongs to the Russell 2000 in year t.

The paper estimates this first stage with first-, second-, and third-order polynomials in log market capitalization and uses year fixed effects and firm-level clustered standard errors.

Interpretation. The first stage asks whether index assignment really changes passive ownership after holding constant the size variable that determines index membership. The answer is yes, strongly.

3.4 Why the instrument uses index assignment, not actual Russell weights

This is an important design choice.

- The paper does *not* instrument with actual Russell portfolio weights or ranks.
- Russell's actual within-index weights depend on float adjustment and other features tied to liquidity and inside ownership.
- Those features could directly affect governance and performance, which would make actual weights a problematic instrument.

Interpretation. Using index assignment rather than exact portfolio weight keeps the instrument closer to an exogenous discontinuity induced by index construction rather than by endogenous features of the firm.

3.5 Why there are no firm fixed effects

- Only a relatively small fraction of firms switch indexes during the sample period.
- The outcomes the paper studies—board structure, takeover defenses, voting rights, and longer-term performance—are likely to respond to sustained ownership differences rather than one-year transitory shocks.
- Firm fixed effects would largely remove that sustained cross-sectional variation and could therefore obscure the economically relevant effect.

Interpretation. The paper is deliberately using long-lived differences in passive ownership, not trying to identify the effect from the handful of firms that bounce across the cutoff.

3.6 Outcome-specific specifications

The same IV framework is applied to a wide set of outcomes.

- Governance outcomes such as poison-pill removal, special-meeting rights, and dual-class shares are estimated as standardized linear-probability-style outcomes.
- Voting outcomes use the average support for management proposals and for shareholder governance proposals.
- Performance outcomes mainly use ROA; the strongest ROA results appear when the authors add controls for firms that just moved between indexes, because performance effects likely materialize only after ownership has had time to matter.

Interpretation. The empirical design is simple and uniform. What changes across tables is not the identification strategy, but the corporate outcome placed on the left-hand side.

3.7 Alternative ownership measures

The paper also re-runs the design with institution-level 13F data.

- "Quasi-index" ownership, in the Bushee classification, is used as an alternative passive-ownership proxy.
- The results are again positive for board independence and ROA, positive for poison-pill removal and special-meeting rights, negative for dual-class shares, and negative for management-proposal support.

Interpretation. The central findings are about passive investors broadly, not just about one narrow mutual-fund classification exercise.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline estimate

The identifying comparison is between firms just on opposite sides of the Russell 1000/2000 threshold.

- These firms are very similar in size.
- Yet because of index mechanics, firms near the top of the Russell 2000 receive much larger passive benchmark weights than firms near the bottom of the Russell 1000.
- After flexibly controlling for market capitalization and adding float controls and year fixed effects, the remaining discrete jump in passive ownership around the cutoff is treated as exogenous.

Interpretation. The paper is essentially asking: among otherwise similar firms near the cutoff, what happens when passive funds own more of one firm because of benchmark design rather than because the firm "asked" for that ownership?

4.2 Relevance: the instrument is strong

The first stage is one of the cleanest parts of the paper.

- Table 2 shows that Russell 2000 assignment increases passive mutual fund ownership by 1.086 percentage points.
- Table 3 shows this corresponds to about 0.47–0.51 standard deviations of passive ownership.
- The effect is statistically significant at the 1% level in every first-stage specification.
- There is no corresponding jump in active mutual fund ownership or unclassified mutual fund ownership.

Interpretation. The discontinuity is specific to passive investors, exactly as the institutional design predicts. That is a major credibility point.

4.3 Exclusion restriction: why the paper thinks index membership only matters through passive ownership

The central exclusion restriction is that, conditional on market capitalization and float, Russell 2000 membership affects governance and performance only through passive ownership.

The paper makes this more plausible by:

- restricting attention to firms close to the cutoff;
- flexibly controlling for end-of-May market capitalization with first-, second-, and third-order polynomials;
- including float-adjusted market value controls;
- including year fixed effects;
- avoiding the use of actual Russell weights as an instrument.

Interpretation. The paper cannot literally prove the exclusion restriction, but it does a lot to make the treated and control firms look like near-neighbors except for passive ownership.

4.4 Alternative explanations and how the paper addresses them

1. Maybe active investors are the true driver.

- The data show no jump in active mutual fund ownership at the cutoff.
- This makes it hard to interpret the results as reflecting activism by active mutual funds rather than passive funds.

2. Maybe unobserved size-related variables drive the result.

- Results are robust to alternative ways of measuring market capitalization using Russell and Compustat data.
- Results are robust to adding industry controls, past return controls, liquidity controls, and controls for recent index switchers.

3. Maybe the result depends on one specific bandwidth choice.

- The paper shows similar estimates across bandwidths from 100 to 500 firms around the threshold.

4. Maybe there is a generic discontinuity at any arbitrary size threshold.

- Placebo tests using fake thresholds within the Russell 1000 or within the Russell 2000 find no comparable effects.

5. Maybe "fallen angels" and "rising stars" are disproportionately on one side.

- The paper finds no meaningful evidence that lagged returns or big market-cap changes around reconstitution drive the results.

6. Maybe the ISS subsample is selectively chosen.

- The paper checks that the results are not obviously due to selection into the ISS sample, especially for dual-class firms.

4.5 Why the activist-story alternative is not the main mechanism

One concern is that passive ownership might matter only because it makes it easier for hedge fund activists or acquirers to coordinate support.

The paper argues against this being the main story.

- A one-standard-deviation increase in passive ownership is associated with a 1.6–2.0 percentage point *decline* in the likelihood of hedge fund activism.
- There is also no evidence that greater passive ownership raises the chance of a takeover event.

Interpretation. The evidence fits a substitution story better than a complement story: passive funds themselves appear to be doing some governance work, reducing the need for outside activists.

4.6 What the paper can and cannot distinguish

What it establishes fairly well.

- Passive funds can affect governance through durable ownership and proxy voting.
- The effects are economically meaningful on governance structures that passive funds publicly say they care about.
- These changes are associated with better longer-term firm performance.

What it does not fully settle.

- It does not prove that every governance change encouraged by passive investors is optimal for every firm.
- It does not say whether passive funds tailor governance recommendations firm by firm or apply a scalable "best practices" checklist.
- It does not show that passive investors become deeply involved in highly firm-specific issues such as capital structure or pay design.

4.7 Relation to the literature

The paper sits at the intersection of three literatures.

- **Institutional ownership and governance:** it extends work showing that institutions affect governance and policies.
- **Activism and monitoring:** it shows that investors do not need activist trading strategies to matter; large passive owners can influence firms through voting and engagement.
- **Russell-index identification papers:** it uses the Russell cutoff specifically to isolate passive ownership, rather than institutional ownership more generally.

Its distinctive contribution is to show that passive institutions can be important monitors even when their investment mandates are mechanically index-based.

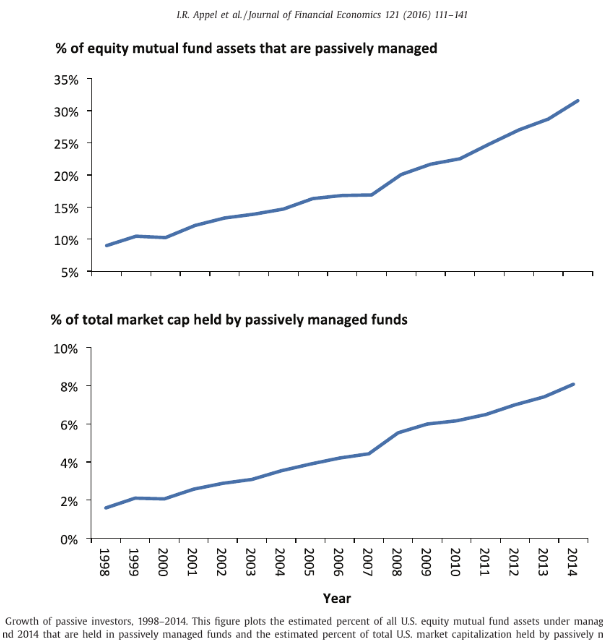
5 Tables and figures

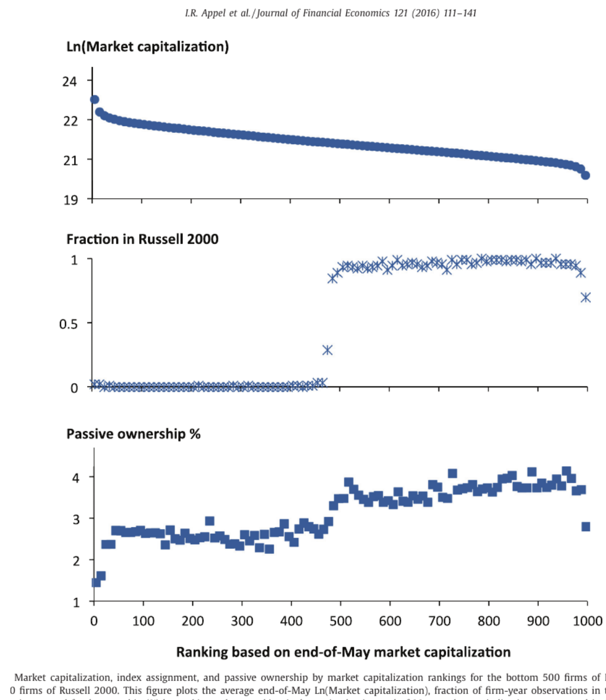
5.1 Figure 1: passive investors become quantitatively important

Read:

- The share of equity mutual fund assets in passive funds rises steadily from the late 1990s onward.
- The share of total U.S. market capitalization held by passive mutual funds also rises sharply.
- By 2014, passive funds are large enough that they can no longer be dismissed as a minor ownership category.

Economic message. The paper's motivation is structural, not anecdotal. Passive ownership became large enough that whether these investors monitor firms is a first-order corporate-governance question.





5.2 Figure 3 and Tables 2–3: the Russell cutoff creates a clean first stage

Read:

- The top panel of Figure 3 shows that market capitalization changes smoothly around the cutoff.
- The middle panel shows that the probability of landing in the Russell 2000 jumps sharply at the cutoff.
- The bottom panel shows a visible jump in passive mutual fund ownership at the same point.
- Table 2 quantifies this: Russell 2000 assignment raises passive mutual fund ownership by about 1.086 percentage points.
- Table 3 shows this is about 0.47–0.51 standard deviations of passive ownership, depending on the polynomial order.
- The same tables show no comparable jump in active mutual fund ownership.

Economic message. This is the heart of identification. The paper gets a discontinuity in passive ownership without a discontinuity in firm size or in active ownership.

Table 2
Impact of index assignment on mutual fund ownership.
This table reports estimates of a regression of mutual fund holdings on an indicator for membership in the Russell 2000 index plus additional controls. Specifically, we estimate

$$Ownership\%_{it} = \eta + \lambda R2000_{it} + \sum_{n=1}^{N} \chi_n ((Ln(Mktcap_{it}))^n + \sigma Ln(Float)_{it} + \delta_t + u_{it},$$

where $R2000_{it}$ is a dummy variable equal to one if stock i is in the Russell 2000 index at end of June in year t, $Mktcap_{it}$ is the CRSP market value of equity of stock i measured at May 31 in year t, N is the polynomial order we use to control for $Ln(Mktcap_{it})$, $Float_{it}$ is the float-adjusted market value of equity (provided by Russell) at June 30 in year t, and δ_t are year fixed effects. $Ownership\%_{it}$ measures mutual fund ownership (in percent) for stock i at the end of September in year t. In this table we use four different definitions for *Ownership%* for stock i: (1) the percentage of shares outstanding owned by all mutual funds (from S12 filings); (2) the percentage of shares outstanding owned by "passive" funds; (3) the percentage of shares outstanding owned by "active" mutual funds; and (4) the percentage of shares outstanding owned by "unclassified" mutual funds. The mutual fund classifications are defined in Section 2.1 of the text. The sample consists of the top 250 firms in the Russell 2000 index and bottom 250 firms of the Russell 1000 index (i.e., bandwidth = 250) for which we obtain holdings data from Thomson Reuters Mutual Fund Holdings Database and which we match with data from the monthly CRSP file. The model is estimated over the 1998–2006 period using a polynomial order control for *Ln(Mktcap)* of $N = 3$. Standard errors, ε, are clustered at the firm level and reported in parentheses. The symbols * and *** indicate significance at the 10% and 1% levels, respectively.

Dependent variable =	Percent of firm's common shares held by:			
	All mutual funds	Passive	Active	Unclassified
	(1)	(2)	(3)	(4)
R2000	1.216*	1.086***	0.118	0.012
	(0.662)	(0.067)	(0.604)	(0.135)
Bandwidth	250	250	250	250
Polynomial order, *N*	3	3	3	3
Float control	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
# of firms	1,654	1,654	1,654	1,654
Observations	4,415	4,415	4,415	4,415
R-squared	0.21	0.62	0.12	0.09

Table 3
First-stage estimation for ownership by passively managed funds.
This table reports estimates of our first-stage regression of passive ownership onto an indicator for membership in the Russell 2000 index plus additional controls. Specifically, we estimate

$$Passive\%_{it} = \eta + \lambda R2000_{it} + \sum_{n=1}^{N} \chi_n (Ln(Mktcap_{it}))^n + \sigma Ln(Float)_{it} + \delta_t + u_{it},$$

where $R2000_{it}$ is a dummy variable equal to one if stock i is in the Russell 2000 index at end of June in year t, $Mktcap_{it}$ is the CRSP market value of equity of stock i measured at May 31 in year t, $Float_{it}$ is the float-adjusted market value of equity (provided by Russell) at June 30 in year t, and δ_t are year fixed effects. $Passive\%_{it}$ is the percentage of shares outstanding owned by passively managed mutual funds, as defined in Section 2.1 of the text, for stock i at the end of September in year t scaled by its sample standard deviation. The data consist of firms in the two Russell indexes for which we obtain holdings data from Thomson Reuters Mutual Fund Holdings Database and which we match with data from the monthly CRSP file. The model is estimated over the 1998–2006 period using a bandwidth of 250 firms around the Russell 1000/2000 threshold, and polynomial order controls for *Ln(Mktcap)* of $N = 1$, 2, and 3. Standard errors, ε, are clustered at the firm level and reported in parentheses. *** indicates significance at the 1% level.

Dependent variable =	Passive % scaled by its sample standard deviation		
	(1)	(2)	(3)
R2000	0.505***	0.512***	0.473***
	(0.028)	(0.028)	(0.029)
Bandwidth	250	250	250
Polynomial order, *N*	1	2	3
Float control	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
# of firms	1,654	1,654	1,654
Observations	4,415	4,415	4,415
R-squared	0.61	0.62	0.62

5.3 Table 4: passive ownership is associated with more independent boards

Read:

- A one-standard-deviation increase in passive ownership is associated with a 0.65–0.76 standard deviation increase in the share of independent directors.
- The result is statistically significant at the 1% level across polynomial specifications.
- The paper also reports that the effect is stronger before the 2003 exchange reforms that tightened board-independence requirements.

Economic message. Passive investors matter most where a governance margin is live. Before exchange rules forced more independent boards, passive investors appear to have pushed firms toward that structure.

Table 4
Ownership by passive investors and board independence.
This table reports estimates of our instrumental variable estimation used to identify the effect of ownership by passive investors on board independence. Specifically, we estimate

$$Y_{it} = \alpha + \beta\, Passive\%_{it} + \sum_{n=1}^{N} \theta_n (Ln(Mktcap_{it}))^n + \gamma\, Ln(Float)_{it} + \delta_t + \varepsilon_{it},$$

where Y_{it} is the percentage of independent directors on the board of firm i in year t (from Riskmetrics) scaled by its sample standard deviation, $Passive\%_{it}$ is the percentage of shares outstanding owned by passively managed funds (as defined in the text) for stock i at the end of September in year t scaled by its sample standard deviation, $Mktcap_{it}$ is the CRSP market value of equity of stock i measured at May 31 in year t, $Float_{it}$ is the float-adjusted market value of equity (provided by Russell) at June 30 in year t, and δ_t are year fixed effects. We instrument *Passive%* in the above estimation using $R2000_{it}$, an indicator equal to one if firm i is part of the Russell 2000 index in year t. The data consist of firms in the two Russell indexes for which we obtain holdings data from Thomson Reuters Mutual Fund Holdings Database and which we match with data from the monthly CRSP file. The model is estimated over the 1998–2006 period using 250 firms around the Russell 1000/2000 threshold, and polynomial order controls for *Ln(Mktcap)* of $N = 1$, 2, and 3. Standard errors, ε, are clustered at the firm level and reported in parentheses. *** indicates significance at the 1% level.

Dependent variable =	Independent director %		
	(1)	(2)	(3)
Passive %	0.729***	0.762***	0.654***
	(0.160)	(0.162)	(0.159)
Bandwidth	250	250	250
Polynomial order, *N*	1	2	3
Float control	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
# of firms	1,082	1,082	1,082
Observations	2,871	2,871	2,871

Table 6
Ownership by passive investors and takeover defenses.
This table reports estimates of our instrumental variable estimation used to identify the effect of institutional ownership by passive investors on takeover defense outcomes. Specifically, we estimate

$$Y_{it} = \alpha + \beta\, Passive\%_{it} + \sum_{n=1}^{N} \theta_n (Ln(Mktcap_{it}))^n + \gamma\, Ln(Float)_{it} + \delta_t + \varepsilon_{it},$$

where Y_{it} is the governance variable for firm i in year t scaled by its sample standard deviation, $Passive\%_{it}$ is the percentage of shares outstanding owned by passively managed mutual funds (as defined in Section 2.1 of the text) for stock i at the end of September in year t scaled by its sample standard deviation, $Mktcap_{it}$ is the CRSP market value of equity of stock i measured at May 31 in year t, $Float_{it}$ is the float-adjusted market value of equity (provided by Russell) at June 30 in year t, and δ_t are year fixed effects. The governance variables investigated in this table, from Shark Repellent (FactSet) and Riskmetrics, are: an indicator for either the withdrawal or expiration (without renewal) of a poison pill in year t, and an indicator for there being fewer restrictions on shareholders' ability to call a special meeting in year t. We instrument *Passive%* in the above estimation using $R2000_{it}$, an indicator equal to one if firm i is part of the Russell 2000 index in year t. The data consist of firms in the two Russell indexes for which we obtain holdings data from Thomson Reuters Mutual Fund Holdings Database and which we match with data from the monthly CRSP file. The model is estimated over the 1998–2006 period using a bandwidth of 250 firms around the Russell 1000/2000 threshold and first-, second-, and third-order polynomial controls for *Ln(Mktcap)*. Standard errors, ε, are clustered at the firm level and reported in parentheses. *** indicates significance at the 1% levels.

Dependent variable =	Poison pill removal			Greater ability to call special meeting		
	(1)	(2)	(3)	(4)	(5)	(6)
Passive %	0.176***	0.181***	0.203***	0.304***	0.310***	0.341***
	(0.0647)	(0.0650)	(0.0741)	(0.0999)	(0.108)	(0.114)
Bandwidth	250	250	250	250	250	250
Polynomial order, *N*	1	2	3	1	2	3
Float control	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
# of firms	1,164	1,164	1,164	1,050	1,050	1,050
Observations	2,957	2,957	2,957	1,858	1,858	1,858

Table 7
Ownership by passive investors and dual class share structures.
This table reports estimates of our instrumental variable estimation used to identify the effect of passive investors on the likelihood of dual class shares. Specifically, we estimate

$$Y_{it} = \alpha + \beta\, Passive\%_{it} + \sum_{n=1}^{N} \theta_n (Ln(Mktcap_{it}))^n + \gamma\, Ln(Float)_{it} + \delta_t + \varepsilon_{it},$$

where Y_{it} is an indicator equal to one if firm i has dual class shares in year t according to Riskmetrics scaled by its sample standard deviation, $Passive\%_{it}$ is the percentage of shares outstanding owned by passively managed mutual funds (as defined in Section 2.1 of the text) for stock i at the end of September in year t scaled by its sample standard deviation, $Mktcap_{it}$ is the CRSP market value of equity of stock i measured at May 31 in year t, $Float_{it}$ is the float-adjusted market value of equity (provided by Russell) at June 30 in year t, and δ_t are year fixed effects. We instrument *Passive%* in the above estimation using $R2000_{it}$, an indicator equal to one if firm i is part of the Russell 2000 index in year t. The data consist of firms in the two Russell indexes for which we obtain holdings data from Thomson Reuters Mutual Fund Holdings Database and which we match with data from the monthly CRSP file. The model is estimated over the 1998–2006 period using a bandwidth of 250 firms around the Russell 1000/2000 threshold, and polynomial order controls for *Ln(Mktcap)* of $N = 1$, 2, and 3. Standard errors, ε, are clustered at the firm level and reported in parentheses. *** indicates significance at the 1% level.

Dependent variable =	Indicator for dual class shares		
	(1)	(2)	(3)
Passive %	−0.886***	−1.031***	−1.005***
	(0.179)	(0.167)	(0.181)
Bandwidth	250	250	250
Polynomial order, *N*	1	2	3
Float control	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
# of firms	1,050	1,050	1,050
Observations	1,858	1,858	1,858

5.4 Tables 6 and 7: passive ownership is associated with fewer takeover defenses and fewer dual-class structures

Read:

- A one-standard-deviation increase in passive ownership raises the likelihood of poison-pill removal by about 3.3–3.8 percentage points.
- The same increase raises the likelihood of reducing restrictions on special meetings by about 2.4–2.7 percentage points.
- Passive ownership is also associated with about a one-standard-deviation decrease in the likelihood of having a dual-class share structure.

Economic message. The governance changes line up tightly with the issues passive institutions publicly said they cared about: fewer entrenchment devices and more equal shareholder voting rights.

5.5 Table 8: passive investors appear to use voice

Read:

- Higher passive ownership is associated with a 0.73–0.78 standard deviation decline in support for management proposals.
- Higher passive ownership is associated with a 0.49–0.65 standard deviation increase in support for shareholder governance proposals.
- The paper finds little evidence that these vote effects come from a changing mix of proposal types.

Economic message. Passive investors do not look passive in the voting booth. They appear to challenge management more and to support governance-oriented shareholder initiatives more.

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Table 8
Ownership by passive investors and shareholder support for proposals.
This table reports estimates of our instrumental variable estimation to identify the effect of passive investors on shareholder support for management proposals and shareholder-initiated governance proposals. Specifically, we estimate

$$Y_{it} = \alpha + \beta\, Passive\%_{it} + \sum_{n=1}^{N} \theta_n (Ln(Mktcap_{it}))^n + \gamma\, Ln(Float)_{it} + \delta_t + \varepsilon_{it},$$

where Y_{it} is either the average percentage of shareholders that vote along with management proposals at annual meetings for firm i in year t (from Riskmetrics) or the average percentage of shareholders that vote in support of a shareholder-initiated governance proposal for firm i in year t (from Riskmetrics) each scaled by their sample standard deviation, $Passive\%_{it}$ is the percentage of shares outstanding owned by passively managed mutual funds (as defined in Section 2.1 of the text) for stock i at the end of September in year t scaled by its sample standard deviation, $Mktcap_{it}$ is the CRSP market value of equity of stock i measured at May 31 in year t, $Float_{it}$ is the float-adjusted market value of equity (provided by Russell) at June 30 in year t, and δ_t are year fixed effects. We instrument *Passive%* in the above estimation using $R2000_{it}$, an indicator equal to one if firm i is part of the Russell 2000 index in year t. The data consist of firms in the two Russell indexes for which we obtain holdings data from Thomson Reuters Mutual Fund Holdings Database and which we match with data from the monthly CRSP file. The model is estimated over the 1998–2006 period using a bandwidth of 250 firms around the Russell 1000/2000 threshold, and polynomial order controls for *Ln(Mktcap)* of $N = 1$, 2, and 3. Standard errors, ε, are clustered at the firm level and reported in parentheses. The symbols *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Dependent variable =	Management proposal support %			Governance proposal support %		
	(1)	(2)	(3)	(4)	(5)	(6)
Passive %	−0.783***	−0.745***	−0.734***	0.492**	0.649*	0.622*
	(0.180)	(0.179)	(0.231)	(0.247)	(0.348)	(0.336)
Bandwidth	250	250	250	250	250	250
Polynomial order, *N*	1	2	3	1	2	3
Float control	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
# of firms	775	775	775	127	127	127
Observations	1,288	1,288	1,288	202	202	202

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Table 10
Ownership by passive investors and firms' return on assets.
This table reports estimates of our instrumental variable estimation used to identify the effect of ownership by passive institutional investors on firms' performance, as measured using firms' return on assets (ROA). Specifically, we estimate

$$Y_{it} = \alpha + \beta\, Passive\%_{it} + \sum_{n=1}^{N} \theta_n (Ln(Mktcap_{it}))^n + \gamma\, Ln(Float)_{it} + \delta_t + \varepsilon_{it},$$

where Y_{it} is the *ROA* for firm i in year t scaled by its sample standard deviation, $Passive\%_{it}$ is the percentage of shares outstanding owned by passively managed mutual funds (as defined in Section 2.1 of the text) for stock i at the end of September in year t scaled by its sample standard deviation, $Mktcap_{it}$ is the CRSP market value of equity of stock i measured at May 31 in year t, $Float_{it}$ is the float-adjusted market value of equity (provided by Russell) at June 30 in year t, and δ_t are year fixed effects. We instrument *Passive%* in the above estimation using $R2000_{it}$, an indicator equal to one if firm i is part of the Russell 2000 index in year t. The specification in columns 1–3 is the same as in earlier tables, but in columns 4–6, we add two additional controls to the specification: an indicator that equals one for firms that are in the Russell 2000 index in year t but were in the Russell 1000 in year $t-1$, and an indicator that equals one for firms that are in the Russell 1000 index in year t but were in the Russell 2000 index in year $t-1$. The data consist of firms in the two Russell indexes for which we obtain holdings data from Thomson Reuters Mutual Fund Holdings Database and which we match with data from the monthly CRSP file. The model is estimated over the 1998–2006 period using a bandwidth of 250 firms around the Russell 1000/2000 threshold and first-, second-, and third-order polynomial controls for *Ln(Mktcap)*. Standard errors, ε, are clustered at the firm level and reported in parentheses. *** indicates significance at the 1% level.

Dependent variable =	ROA					
	(1)	(2)	(3)	(4)	(5)	(6)
Passive %	−0.028	−0.015	0.035	0.304***	0.310***	0.414***
	(0.098)	(0.093)	(0.105)	(0.111)	(0.106)	(0.121)
Bandwidth	250	250	250	250	250	250
Polynomial order, *N*	1	2	3	1	2	3
Float control	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Controls for movers	No	No	No	Yes	Yes	Yes
# of firms	1,600	1,600	1,600	1,600	1,600	1,600
Observations	4,291	4,291	4,291	4,291	4,291	4,291

acquisitions, cash levels, and payout policies (Jensen, 1986; La Porta, Lopez-de-Silanes, and Shleifer, 2000). To examine whether ownership by passive mutual funds is associated

possibility of a selection bias around the Russell 1000/2[illegible] threshold, particularly for the subsample of observati[illegible] covered by ISS databases.

5.6 Table 10: longer-term performance improves

Read:

- In the baseline ROA specification, the coefficients are small and insignificant.
- Once the paper controls for firms that recently switched indexes, the coefficients become positive and significant.
- In the stronger specifications, a one-standard-deviation increase in passive ownership raises ROA by about 0.30–0.41 standard deviations.

Economic message. Governance changes appear to translate into better operating performance, but not immediately. The effect looks like a longer-run ownership effect rather than a contemporaneous mechanical consequence of index assignment.

6 Short summary

- The paper asks whether passive institutional investors weaken or improve corporate governance.
- It exploits the Russell 1000/2000 cutoff, which generates large discontinuities in passive ownership but not in active mutual fund ownership.
- Using index assignment as an instrument, the authors show that more passive ownership is associated with more independent directors, fewer poison pills, greater shareholder meeting rights, and fewer dual-class structures.
- The mechanism evidence points to *voice*: passive ownership lowers support for management proposals and raises support for governance-related shareholder proposals.
- The paper does not find evidence that passive ownership works by encouraging more hedge fund activism. If anything, activism becomes less likely.
- Longer-term firm performance improves, especially in specifications that allow time for the ownership change to matter.
- The overall lesson is that passive investors may be passive in portfolio construction, but they are not passive as owners.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that passive mutual funds can causally influence firms' governance structures. Using quasi-experimental variation from Russell index assignment, the authors show that greater passive ownership is associated with more independent boards, fewer entrenching provisions, more equal voting rights, more opposition to management proposals, and stronger support for shareholder governance proposals.

7.2 Economic intuition

Passive investors cannot easily discipline managers by selling a stock without deviating from their benchmark, but they often hold large, long-lived stakes and vote every year. That makes scalable governance issues—board composition, takeover defenses, voting rights, and proposal voting—exactly the margins on which passive investors can matter. Their influence comes less from trading and more from persistent ownership plus voice.

7.3 Takeaway

The rise of passive investing does not mechanically imply weaker monitoring. In this paper, the opposite is closer to the truth: passive investors look like engaged owners on broad governance questions, and their growing presence changes who monitors managers in modern public firms.